Evolving Technology & Threats

Final Assignment

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Here is a list of 5 security topics I believe should be addressed for the publishing firm. Following these recommendations will lower the risks involved with using technology in the workplace.

1. **Proper Cyber Security Training**

   Human error is the root of most network security flaws. If staff are not practicing safe usage of the network, it negates most of the security features implemented in the system. Training should be regular and updated to account for how technology is evolving. Here are some baseline topics that staff should be educated on:

   - <u>Secure Access Methods</u>
     - Using weak/default passwords, with limited or no other authentication methods leaves the network extremely vulnerable for intrusion. These passwords can easily be guessed/cracked, allowing a bad actor to have a doorway into the network. I would recommend using passphrases (string of words), that are not used anywhere else & incorporate special characters, numbers and capital letters. Implementing an additional layer of authentication (MFA) is also highly recommended. This could be a code sent by text./email, or biometrics.
   - <u>Social Engineering Awareness</u>
     - The methods scammers use continue to grow more sophisticated. They can spoof their emails to look legitimate, target specific

employees/departments with relevant content in phishing emails, etc. Employees should be critical of every email they receive. Links in emails should rarely be clicked and should be verified before doing so. Any request for information that is deemed sensitive should be verified (ex. call the sender to make sure it is a legitimate request). Make sure to check the email address of the sender, and if it seems suspicious, contact the security team before proceeding.

- Proper Use of Data and Devices
    - Staff should be informed on the risk of using devices and accessing data in insecure ways. Company data & assets should only be used within the company's network/VPN. Using external networks is risky, as the security team is not able to ensure the safety of the network.

2. Securing the Network

A network should be built and configured with security in mind, and security methods should be implemented to a degree deemed adequate for the company's assets. Security practices should be evaluated regularly and modified to reflect the current threats/vulnerabilities. This covers a wide scope of risks that could lead to an attacker breaching the network. Here are some recommendations for security implementations to strengthen the network:

- The network should be assessed and segmented, ensuring critical components (ex. servers) are not affected if other, less critical devices are breached.
- Using a VPN with strong security is recommended.

- Resources/Data should be available based on the principal of least privilege; deny everything by default and only allow use of what's necessary.
- Regularly back up systems to have fail safety
- Research and ensure devices/software are secure before adding them into the network. Avoid unnecessary applications and devices.
- Use firewalls, antivirus software, IDS systems, etc. to a degree deemed appropriate. These need to be configured to detect and block anything that is unneeded.

3. Implementing AI

Generative AI models have the power to increase productivity greatly and can help simplify many tasks. The publishing industry has been adopting AI to help with processes like text analysis, marketing and data analytics. Users should be cautious of what information they provide to AI models, as there are concerns for how this data will be used and stored. It is also important to recognize that AI could be trained maliciously to give potentially harmful responses. If AI is going to be used, here are some practices to help mitigate the associated risks:

- Refrain from submitting anything that is deemed sensitive and confidential
- Research AI tools before using them. Ensure a model does not have any known vulnerabilities or poor data handling processes. The organization should have a policy in place on what AI tools are authorized based on this, and what is acceptable usage of the tool.
- Keep in mind that information outputted by AI may not be accurate; double check outputs before using them.

- AI outputs may use copyrighted data; check the sources of outputs to ensure you do not violate intellectual property. Make sure your inputs do not contain any information that can violate your organizations IP usage.

4. Using Cloud Services

Cloud services (ex. Microsoft 365, AWS servers) give an organization flexible to implement what they need, when they need it, ensuring scalability and lowering costs. Organizations do not own the cloud services they implement; they licence a piece of it for usage. This leads to concerns as the security of the service is out of the control of the organization, therefore it is important to understand the risks before implementing them. Cloud services can be attacked and breached, which can have impacts on the user's data and functionality. Here are some ways to reduce the risks of using cloud platforms:

- The organization should research and develop a policy on what cloud services are deemed safe to use, and what practices they are to be used for.
- Properly configure cloud services to maximize the security on your end. This should be based on a zero-trust model (assume no trust by default, manually allow requests where possible)
- Use secure methods for accessing cloud services. Restrict usage based on the principal of least privilege. Use strong passwords and multi-factor authentication where possible.
- Segment cloud platforms from other critical systems in the network.
- Use encryption for your data, both at rest and during transit with cloud platforms.

- Ensure you have backups and response plans in place in case of failure/breach of a cloud service.
- Limit the usage of sensitive data as much as possible.
- Keep cloud software up to date with security patches.

5. Maintaining a Secure Network

A network needs continuous work to ensure the security practices are sufficient and working as intended. If a network is neglected, software could become outdated and vulnerable, breaches could occur without notice, etc. Here are some key strategies to keep things secure over time:

- Have proper monitoring tools that log any security concerns. These logs should be monitored, analyzed and addressed.
- Regularly audit software to ensure what is on the network is secure and up to date. Check for any new vulnerabilities that may exist in software. Regularly check for any updates and ensure updates do not have any zero-day exploits before deployment.
- Upgrade old devices and software if they are out of date and are insufficiently secure.
- Regularly back up critical systems and data.
- Perform risk assessments of assets regularly. Ensure you are aware of the risks associated with everything on your network.
- Have incident response plans in place. Make sure you know how to act in case of emergency. This could include quarantining devices/network segments, switching over to back up services, and restoring things from backups.

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